Process Flow Map & KOP – Jai Shri Ram Furniture by ModuCraft

ModuCraft
JAI SHRI RAM FURNITURE

PROCESS FLOW MAP

START

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Material Selection (HDHMR / Plywood / Board)

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Thickness Finalization (7.5 MM/ 11 MM/ 16 MM / 16.75 MM/ 18 MM)

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Surface Cleaning (Dust-Free)

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Adhesive Application (Pidilite Products)

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Laminate / Acrylic Application

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Cold Press (50 Ton)

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Beam Saw Cutting

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Edge Banding

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Process Flow Map & KOP – Jai Shri Ram Furniture by ModuCraft

Multi-Boring

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CNC Routing (Jhiri Cuts)

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Packaging (Corrugated + Shrink Wrap)

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READY FOR INSTALLATION

DETAILED PROCESS (STEP-WISE KOP)

1. Material Selection

Selection of base material such as HDHMR, Plywood, or Block boards is done based on application, load-bearing requirements, moisture exposure, and client budget.

2. Thickness Finalization

Thickness (commonly 16 MM/ 16.75 MM or 18mm) is decided based on structural strength, design requirement, and usage.

3. Surface Preparation

Surfaces must be completely clean and dust-free before adhesive application.

4. Adhesive Application

Adhesive is applied evenly on both surfaces to ensure strong bonding.

5. Laminate / Acrylic Application

Sheets are applied carefully ensuring proper alignment and no bubbles.

6. Cold Press Process

Panels are kept under 50-ton cold press for proper bonding.

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ADVANCED OPERATIONS & KEY PRACTICES

7. Beam Saw Cutting

Precision cutting ensures exact dimensions & it's been executed on Panel Saw/ Beam Saw.

8. Edge Banding

Edges are sealed for durability and aesthetics on Edge Bend Machine.

9. Multi-Boring

Accurate drilling ensures proper fittings & it's been executed on Multi Boring Machine.

10. CNC Routing

Grooving (Jhiri cuts) and design operations is been done on CNC Router.

11. Packaging

Corrugated Box Packaging + shrink wrapping is done for protection.

Key Operational Practices (KOP)

- **Cleanliness during adhesive application is critical -** The surface must be completely dust-free before pasting. Even minor dust particles can lead to bubbles, weak bonding, and surface imperfections.

- **Pasting quality plays a crucial role in product durability -** Proper adhesive application and bonding directly impact the life and finish of the final product.

- **Cold press pressure must be accurate and material-specific -** The pressure applied in the cold press varies depending on the surface material.

    1. Acrylic sheets require different pressure settings
    2. Laminates (Sunmica) require different pressure settings
    3. Incorrect pressure can lead to improper bonding or surface damage.
- **Precision in cutting reduces installation issues -** Accurate cutting on the beam saw ensures proper alignment during installation and minimizes rework at site.
- **Edge banding defines durability and finish quality -** Proper edge banding protects the panel from moisture, damage, and enhances the overall aesthetics.
- **Multi-boring is a key differentiator in modular furniture** - This is one of the major differences between modular furniture and traditional carpenter-made furniture.

Multi-boring ensures:

1. Precise hole positioning
2. Hidden fittings
3. No visible screws or nails
4. Clean and premium finish

- **CNC routing (Jhiri cuts) ensures precision and consistency**

  Machine-based grooving and cuts provide:

  1. High accuracy
  2. Seamless finish
  3. Consistency across all panels. This is not achievable manually.

- **Coordination between design and production teams is essential -** Proper communication ensures that designs are practical, executable, and error-free.
- **Packaging is critical to protect product quality** - Proper use of corrugated sheets and shrink wrapping ensures that the product reaches the site without scratches or damage.